

University Of Engineering and Technology, Lahore
Computer Engineering Department

Course Name: Fundamentals of Programming & Data Science	Course Code: CMPE-112L
Assignment Type: Lab Report	Dated: July 1, 2026
Semester: 1st Semester	Session: Spring 2026
Lab/Project/Assignment #: 3	CLOs to be covered: CLO1
Lab Title: While Loops	Teacher Name: Hafiz Muhammad Mubashar
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Lab Evaluation:

CLO1	Apply appropriate programming techniques to create executable programs to solve well defined problems.					
	Level1	Level2	Level3	Level4	Level5	Level6
(5)						
Total						/5

Rubrics for Current Lab Evaluation

Scale	Marks	Level	Rubric
Excellent	5	L1	Submitted all lab tasks, BONUS task, have good

			understanding.
Very Good	4	L2	Submitted the lab tasks but have good understanding
Good	3	L3	Submitted the lab tasks but have weak understanding.
Basic	2	L4	Submitted the lab tasks but have no understanding.
Barely Acceptable	1	L5	Submitted only one lab task.
Not Acceptable	0	L6	Did not attempt

LAB No 3

Lab Goals / Objectives:

- Understand the concept and syntax of while loops in Python
- Differentiate between for loops and while loops
- Learn to use while loops for condition-based iteration
- Apply while loops with user-defined functions
- Understand function definition, parameters, and return values
- Use while loops for mathematical computations

Equipment Required:

- Computer System with Python 3.x installed
- Text Editor or IDE (PyCharm, VS Code, or IDLE)
- Operating System: Windows / Linux / macOS

Theory:

1. The while Loop in Python

A while loop repeatedly executes a block of code as long as a given condition is True. It is used when the number of iterations is not known in advance.

```
while condition:  
    # Code block to execute
```

The condition is evaluated before each iteration. If it is True, the loop body executes; if False, the loop terminates.

2. For Loop vs While Loop

- For loop: Used when the number of iterations is known or when iterating over a sequence
- While loop: Used when the number of iterations depends on a condition

3. User-Defined Functions

A function is a reusable block of code that performs a specific task. Functions are defined using the def keyword:

```
def function_name(parameters):  
    # Function body  
    return value
```

Functions can accept parameters (arguments) and return values using the return statement.

4. Function Parameters and Arguments

- Parameters are variables listed in the function definition
- Arguments are the actual values passed to the function when called
- Functions can have multiple parameters separated by commas

5. The return Statement

The return statement exits a function and optionally passes a value back to the caller:

```
def add(a, b):  
    return a + b
```

6. Factorial Calculation

The factorial of a number n (denoted as $n!$) is the product of all positive integers less than or equal to n :

$$n! = 1 \times 2 \times 3 \times \dots \times n$$

7. Permutation and Combination

Permutation $P(n,r) = n! / (n-r)!$ represents the number of ways to arrange r items from n items.

Combination $C(n,r) = n! / (r! \times (n-r)!)$ represents the number of ways to choose r items from n items without regard to order.

8. Lambda Functions

A lambda function is a small anonymous function defined with the lambda keyword. It can take any number of arguments but has only one expression:

```
lambda arguments : expression
```

Lambda functions are often used for simple operations that can be written in a single line.

9. Temperature Conversion Formulas

Celsius to Fahrenheit: $F = (C \times 9/5) + 32$

Fahrenheit to Celsius: $C = (F - 32) \times 5/9$

10. GPA Calculation

GPA (Grade Point Average) is calculated by summing the product of grade points and credit hours for all subjects, then dividing by the total credit hours:

$$\text{GPA} = \text{Sum}(\text{Grade Point} \times \text{Credit Hours}) / \text{Total Credit Hours}$$

Lab Work:

Task 1: Print Hello 5 Times

Write a program that uses a while loop to print "Hello" five times.

Code Summary:

This task demonstrates the basic structure of a while loop: a counter variable is initialized, checked against a condition, and incremented on each pass until the condition becomes false.

Key Code Lines:

```
# Task 1: Print Hello 5 Times
count = 1
while count <= 5:
    print("Hello")
    count += 1
```

Output:



```
Run  main x  III x
C:\Users\itsha\AppData\Local\Python\bin\python.exe "C:\Users\itsha\Desktop\All weeks\name\lll.py"
Hello
Hello
Hello
Hello
Hello
Process finished with exit code 0
```

Task 2: Print Numbers 1 to 5

Write a program that uses a while loop to print numbers from 1 to 5.

Code Summary:

This task uses a while loop with a counter variable that is printed and incremented on each iteration until it exceeds 5.

Key Code Lines:

```
# Task 2: Print Numbers 1 to 5
num = 1
while num <= 5:
    print(num)
    num += 1
```

Output:



```
Run  main ×  III ×
C:\Users\itsha\AppData\Local\Python\bin\python.exe "C:\Users\itsha\Desktop\All weeks\name\lll.py"
1
2
3
4
5
Process finished with exit code 0
```

Task 3: Row of Stars

Write a program that uses a while loop to print a fixed row of stars, five times.

Code Summary:

This task repeats a single line of five stars using a while loop controlled by a row counter.

Key Code Lines:

```
# Task 3: Row of Stars
rows = 5
i = 1
while i <= rows:
    print("* * * * *")
    i += 1
```

Output:



```
Run  main ×  III ×
C:\Users\itsha\AppData\Local\Python\bin\python.exe "C:\Users\itsha\Desktop\All weeks\name\lll.py"
* * * * *
* * * * *
* * * * *
* * * * *
* * * * *
Process finished with exit code 0
```

Task 4: Increasing Star Pattern

Write a program that uses a while loop to print an increasing star pattern (1 star, 2 stars, 3 stars...).

Code Summary:

This task multiplies the "*" character by the current counter value on each iteration, producing a triangle of stars using only a while loop.

Key Code Lines:

```
# Task 4: Increasing Star Pattern
rows = 5
i = 1
while i <= rows:
    print("*" * i)
    i += 1
```

Output:

A screenshot of a Python IDE window titled 'main'. The terminal output shows an increasing star pattern: a single line with 1 star, followed by lines with 2, 3, 4, and 5 stars respectively. Below the pattern, it says 'Process finished with exit code 0'. The command prompt shows the file path 'C:\Users\itsha\Desktop\All weeks\name\111.py'.

Task 5: Repeated Digit Pattern

Write a program that uses a while loop to print each number repeated according to its own value (e.g. 1, 22, 333, ...).

Code Summary:

This task converts the counter to a string and repeats it by its own value on each while-loop pass, building a numeric pattern.

Key Code Lines:

```
# Task 5: Repeated Digit Pattern
i = 1
while i <= 5:
```

```
print(i * str(i))  
i += 1
```

Output:



```
Run  main ×  III ×  
C:\Users\itsha\AppData\Local\Python\bin\python.exe "C:\Users\itsha\Desktop\All weeks\name\111.py"  
1  
22  
333  
4444  
55555  
Process finished with exit code 0
```

Conclusions:

- Successfully understood and applied while loops for condition-based iteration
- Learned to define and call user-defined functions with parameters and return values
- Mastered the use of lambda functions for simple operations
- Applied mathematical concepts through programming (factorial, permutation, combination)
- Developed practical utility programs (temperature converter, GPA calculator)
- Understood the difference between for loops and while loops
- Gained experience in modular programming through function decomposition